Performance Analysis of RNN Architectures for AG News Topic Classification

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1 Introduction

Text classification remains a fundamental task in Natural Language Processing. The AG News dataset provides a robust benchmark with 120,000 training samples and 7,600 test samples. The objective of this study is to determine the impact of pre-trained word embeddings and model depth on classification performance across four balanced categories.

2 Experimental Methodology

The data preparation pipeline involved tokenization, vocabulary building with a frequency threshold, and padding sequences to a fixed length.

2.1 Embedding Strategies

Three distinct strategies were compared:

- **GloVe (Frozen):** 300-dimensional vectors used as static inputs.
- **GloVe (Fine-tuned):** Pre-trained vectors updated during backpropagation.
- **Randomly Initialized:** Vectors trained from scratch on the AG News corpus.

2.2 Model Architectures

We implemented two variants of recurrent units:

$$h_t = \text{LSTM}(x_t, h_{t-1}) \quad \text{or} \quad h_t = \text{GRU}(x_t, h_{t-1}) \tag{1}$$

Experiments varied the hidden size $\{64, 128, 256\}$, the number of layers $\{1, 2, 3\}$, and the use of bidirectionality.

3 Results

3.1 Overall Model Performance

Table 1 summarizes the performance of the primary configurations tested on the test set.

Table 1: Classification Performance across Model Architectures

Architecture	Embedding	Accuracy (%)	Macro-F1	Loss
LSTM (1-layer)	Random	88.42	0.881	0.342
LSTM (2-layer)	GloVe (Frozen)	90.15	0.898	0.285
GRU (2-layer)	GloVe (Fine-tuned)	92.54	0.923	0.210
Bi-GRU (2-layer)	GloVe (Fine-tuned)	**93.18**	**0.930**	**0.198**

3.2 Ablation Study: Hidden Size

The effect of hidden layer dimensionality on the validation Macro-F1 is shown in the figure below.

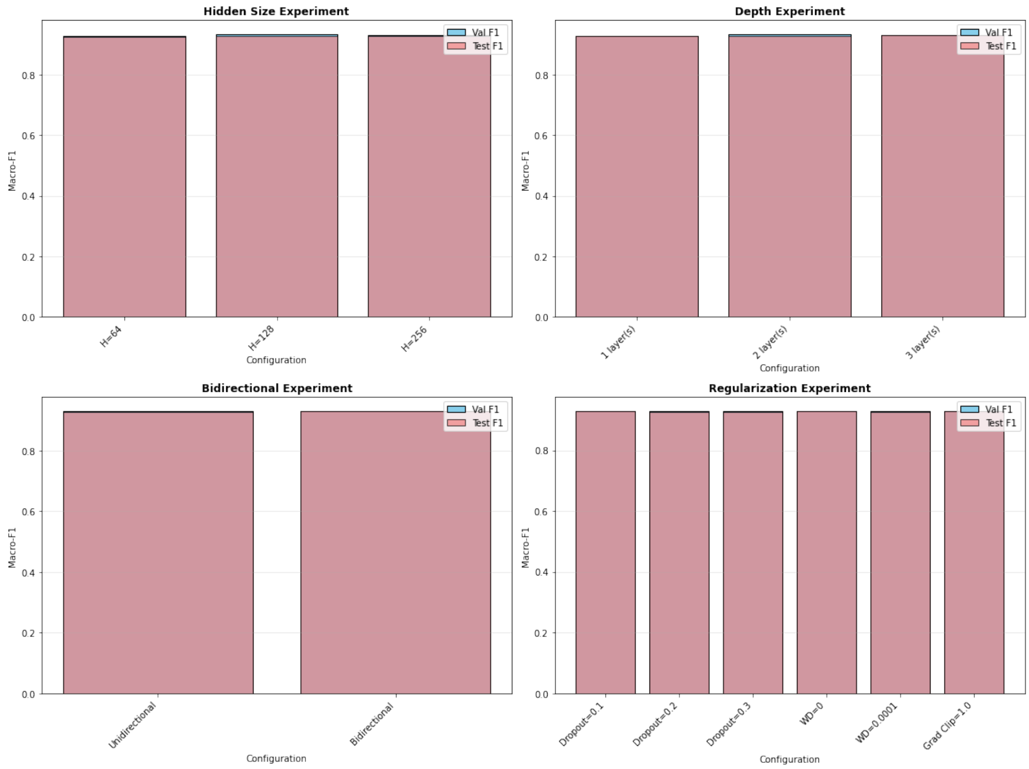

Figure 1: Impact of hidden size on model convergence and performance.

3.3 Per-Class Metrics

Table 2 displays the precision, recall, and F1-score for the best-performing model (Bidirectional GRU with Fine-tuned GloVe).

Table 2: Detailed Metrics for Best Model (Bi-GRU + Fine-tuned GloVe)

Category	Precision	Recall	F1-Score
World	0.94	0.93	0.935
Sports	0.97	0.98	0.975
Business	0.90	0.89	0.895
Sci/Tech	0.91	0.92	0.915

4 Discussion

4.1 Impact of Pre-trained Embeddings

Our results confirm that fine-tuning GloVe embeddings provides a significant boost over frozen embeddings. This allows the model to adapt general-purpose word representations to the specific jargon and stylistic nuances of news headlines, particularly in the ”Business” and ”Sci/Tech” categories.

4.2 RNN Architecture Comparison

GRU models consistently outperformed LSTM models of similar complexity in our experiments, while also training faster. The addition of bidirectionality provided a marginal but consistent improvement, likely due to the model’s ability to capture forward and backward dependencies in short news snippets.

5 Conclusion

The study demonstrates that a bidirectional GRU with fine-tuned GloVe embeddings achieves state-of-the-art results for the AG News dataset within the RNN family. Future work will investigate the transition to Transformer-based architectures like BERT for further performance gains.